

Solutions to Dummit & Foote's Abstract Algebra 3rd Edition

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Preface

This is a collection of solutions to the exercises in Dummit and Foote's *Abstract Algebra*, 3rd edition. These solutions were written by me as I worked through the book, and are intended to serve as a reference for myself and others who are studying abstract algebra. I have made every effort to ensure the correctness of these solutions, but I cannot guarantee that they are free of errors. If you find any mistakes or have suggestions for improvement, please feel free to contact me.

I've attempted to format the solutions in a clear and consistent manner, using LaTeX for typesetting. Each chapter's exercises are included in separate files for better organization if viewing on GitHub. Moreover, solutions to exercises only utilize techniques that are available to an advanced undergraduate student or beginning graduate student, in line with the intended audience of the textbook. As an aside, I've graduated from a bachelor's program in mathematics but have not pursued graduate studies, so my background is primarily at the undergraduate level.

Some suggestions I've been given to improve the guide are citing particularly important exercises that will prove useful in reading subsequent chapters, and providing writeups/discussions for why I choose a particular method of solution (such as functions/homomorphisms that may appear out of nowhere). Most problems are enclosed in the following environment:

Exercise 0.0.1

Example exercise.

However, there are some exercises that I found to be either challenging, interesting, or useful for the development of later material. These exercises are enclosed in a special environment:

(*) Exercise 0.0.2

Example special exercise.

This is to highlight these exercises for future readers who may want to focus on them.

Many thanks to the authors, David S. Dummit and Richard M. Foote, for writing such an excellent textbook that has been a valuable resource for my recreational studies in abstract algebra. I've received requests on Reddit for individuals to assist me in completing this project, but at this time I prefer to work on it independently. However, I welcome feedback and suggestions from anyone who is interested in contributing to the project in the future.

5 Direct and Semidirect Products and Abelian Groups

5.1 Direct Products

Exercise 5.1.1

Show that the center of a direct product is the direct product of the centers:

$$Z(G_1 \times G_2 \times \cdots \times G_n) = Z(G_1) \times Z(G_2) \times \cdots \times Z(G_n)$$

Deduce that the direct product of two groups is abelian if and only if each of the factors is abelian.

Solution. Let $G = G_1 \times G_2 \times \cdots \times G_n$. Let $z = (z_1, z_2, \dots, z_n) \in Z(G)$. Then for any $g = (g_1, g_2, \dots, g_n) \in G$, we have $zg = gz$, or equivalently,

$$(z_1g_1, z_2g_2, \dots, z_ng_n) = (g_1z_1, g_2z_2, \dots, g_nz_n)$$

This implies that $z_i g_i = g_i z_i$ for all i , so $z_i \in Z(G_i)$ for all i . Thus, $z \in Z(G_1) \times Z(G_2) \times \cdots \times Z(G_n)$, and we have shown that $Z(G) \subseteq Z(G_1) \times Z(G_2) \times \cdots \times Z(G_n)$.

Conversely, let $z = (z_1, z_2, \dots, z_n) \in Z(G_1) \times Z(G_2) \times \cdots \times Z(G_n)$. Then for any $g = (g_1, g_2, \dots, g_n) \in G$, we have

$$zg = (z_1g_1, z_2g_2, \dots, z_ng_n) = (g_1z_1, g_2z_2, \dots, g_nz_n) = gz$$

Thus, $z \in Z(G)$, and we have shown that $Z(G_1) \times Z(G_2) \times \cdots \times Z(G_n) \subseteq Z(G)$. ■

Exercise 5.1.2

Let $G_1, G_2, \dots, G_n$ be groups and $G = G_1 \times \cdots \times G_n$. Let I be a proper, nonempty subset of $\{1, \dots, n\}$ and let $J = \{1, \dots, n\} - I$. Define G_I to be the set of elements of G that have the identity G_j in position j for all $j \in J$.

- Prove that G_I is isomorphic to the direct product of the G_i for $i \in I$.
- Prove that G_I is a normal subgroup of G and that $G/G_I \cong G_J$.
- Prove that $G \cong G_I \times G_J$.

Solution.

- Identifying G_i with the subgroup of G defined in Proposition 5.2, let H be the product of the G_i for $i \in I$ in G . Let $\varphi : G_I \rightarrow H$ be the map defined by mapping the non-identity coordinates of G_I to the corresponding coordinates of H . It is clear that φ is a homomorphism. Moreover, φ is surjective by definition of H . Finally, the kernel of φ is the set of all elements of G_I that have the identity in all coordinates, which is just the identity element of G . Thus, φ is an isomorphism from G_I to H , and we are done.
- Consider the map $\varphi : G \rightarrow G_J$ defined by mapping the non-identity coordinates of G to the corresponding coordinates of G_J . It is clear that φ is a homomorphism. Moreover, φ is surjective by definition of G_J . Finally, the kernel of φ is the set of all elements of G that have the identity in all coordinates corresponding to J , which is exactly G_I . Thus, φ induces an isomorphism from G/G_I to G_J , and we are done.
- Any element of $G_I \times G_J$ can be written as the product of an element of G_I and an element of G_J by taking the first and second coordinates, respectively. Thus, the map $\varphi : G_I \times G_J \rightarrow G$ defined by $\varphi((g_i)_{i \in I}, (g_j)_{j \in J}) = (g_1, g_2, \dots, g_n)$ is a bijection. Injectivity is clear, since $\ker \varphi$ is trivial. It is also a homomorphism since the group operation in G is defined coordinate-wise. Thus, φ is an isomorphism from $G_I \times G_J$ to G , and we are done. ■

Exercise 5.1.3

Under the notation of the preceding exercise, let I and K be any disjoint nonempty subsets of $\{1, 2, \dots, n\}$, and let G_I and G_K be the subgroups of G defined above. Prove that $xy = yx$ for all $x \in G_I$ and $y \in G_K$.

Solution. Let $x \in G_I$ and $y \in G_K$. Then x has the identity in all coordinates corresponding to $J = \{1, 2, \dots, n\} - I$, and y has the identity in all coordinates corresponding to $L = \{1, 2, \dots, n\} - K$. Since I and K are disjoint, we have $J \cup L = \{1, 2, \dots, n\}$, so every coordinate of x and y is the identity in at least one of them. Thus, when we compute the product xy , we have

$$xy = (x_1y_1, x_2y_2, \dots, x_ny_n) = (y_1x_1, y_2x_2, \dots, y_nx_n) = yx$$

as desired. ■

Exercise 5.1.4

Let A and B be finite groups, and let p be a prime. Prove that any Sylow p -subgroup of $A \times B$ is of the form $P \times Q$, where $P \in \text{Syl}_p(A)$ and $Q \in \text{Syl}_p(B)$. Prove that $n_p(A \times B) = n_p(A)n_p(B)$. Generalize both of these results to a direct product of any finite number of finite groups (so that the number of Sylow p -subgroups of a direct product is the product of the numbers of Sylow p -subgroups of the factors).

Solution. Let $|A| = p^\alpha m$ and $|B| = p^\beta n$, where $p \nmid m$ and $p \nmid n$. Then $|A \times B| = |A||B| = p^{\alpha+\beta}mn$. Let $R \in \text{Syl}_p(A \times B)$. Then $|R| = p^{\alpha+\beta}$, so R is a Sylow p -subgroup of $A \times B$. Let $\pi_A : A \times B \rightarrow A$ and $\pi_B : A \times B \rightarrow B$ be the projection maps. Then $\pi_A(R)$ is a p -subgroup of A , so $\pi_A(R) \subseteq P$ for some $P \in \text{Syl}_p(A)$. Similarly, $\pi_B(R) \subseteq Q$ for some $Q \in \text{Syl}_p(B)$. Thus, $R \subseteq P \times Q$. Since $|P \times Q| = |P||Q| = p^{\alpha+\beta}$, we must have $R = P \times Q$.

To show $n_p(A \times B) = n_p(A)n_p(B)$, we note that the Sylow p -subgroups of $A \times B$ are precisely the subgroups of the form $P \times Q$, where $P \in \text{Syl}_p(A)$ and $Q \in \text{Syl}_p(B)$. Since there are $n_p(A)$ choices for P and $n_p(B)$ choices for Q , we have $n_p(A \times B) = n_p(A)n_p(B)$. ■

Exercise 5.1.5

Exhibit a nonnormal subgroup of $Q_8 \times Z_4$ (note that every subgroup of each factor is normal).

Solution. Note that Q_8 is non-abelian, so we may use the idea of a diagonal subgroup from [Exercise 3.1.38](#). In particular, let $H = \langle (i, 1) \rangle = \{(1, 0), (i, 1), (-1, 2), (-i, 3)\}$. Then H is a subgroup of $Q_8 \times Z_4$ that is isomorphic to $\langle i \rangle$. However,

$$(j, 0)(i, 1)(j, 0)^{-1} = (ji(-j), 0 + 1 + 0) = (-i, 1)$$

is not in H , so H is not normal in $Q_8 \times Z_4$. ■

Exercise 5.1.6

Show that all subgroups of $Q_8 \times E_{2^n}$ are normal.

Solution. Let $H \leq Q_8 \times E_{2^n}$. Suppose $(a, b) \in H$ and $(x, y) \in Q_8 \times E_{2^n}$. Note that $(a^{-1}, b) \in H$, since every $b \in E_{2^n}$ is its own inverse. Then

$$(x, y)(a, b)(x, y)^{-1} = (xax^{-1}, yby^{-1}) = (xax^{-1}, b)$$

since E_{2^n} is abelian. If $a \in Z(Q_8)$, then $xax^{-1} = a$, so $(x, y)(a, b)(x, y)^{-1} = (a, b) \in H$. If $a \notin Z(Q_8)$, then a is one of i, j, k or their inverses. In all cases, we have xax^{-1} is one of i, j, k or their inverses, so $xax^{-1} \in H$. Thus, $(x, y)(a, b)(x, y)^{-1} \in H$ for all $(a, b) \in H$ and $(x, y) \in Q_8 \times E_{2^n}$, so H is normal in $Q_8 \times E_{2^n}$. ■

Exercise 5.1.7

Let $G_1, G_2, \dots, G_n$ be groups and π be a fixed element of S_n . Prove that the map

$$\varphi_\pi : G_1 \times G_2 \times \cdots \times G_n \rightarrow G_{\pi^{-1}(1)} \times G_{\pi^{-1}(2)} \times \cdots \times G_{\pi^{-1}(n)}$$

defined by

$$\varphi_\pi(g_1, g_2, \dots, g_n) = (g_{\pi^{-1}(1)}, g_{\pi^{-1}(2)}, \dots, g_{\pi^{-1}(n)})$$

is an isomorphism (so that changing the order of the factors in a direct product does not change the isomorphism type).

Solution. Let $\pi \in S_n$ and let φ_π be the map defined in the problem statement. Let $g = (g_1, g_2, \dots, g_n)$ and $h = (h_1, h_2, \dots, h_n)$ be elements of $G_1 \times G_2 \times \cdots \times G_n$. Then

$$\begin{aligned} \varphi_\pi(gh) &= \varphi_\pi(g_1h_1, g_2h_2, \dots, g_nh_n) \\ &= (g_{\pi^{-1}(1)}h_{\pi^{-1}(1)}, g_{\pi^{-1}(2)}h_{\pi^{-1}(2)}, \dots, g_{\pi^{-1}(n)}h_{\pi^{-1}(n)}) \\ &= (g_{\pi^{-1}(1)}, g_{\pi^{-1}(2)}, \dots, g_{\pi^{-1}(n)})(h_{\pi^{-1}(1)}, h_{\pi^{-1}(2)}, \dots, h_{\pi^{-1}(n)}) \\ &= \varphi_\pi(g)\varphi_\pi(h) \end{aligned}$$

Noting that $\pi \in S_n$ is a bijection, consider the map $\varphi_{\pi^{-1}} : G_{\pi^{-1}(1)} \times G_{\pi^{-1}(2)} \times \cdots \times G_{\pi^{-1}(n)} \rightarrow G_1 \times G_2 \times \cdots \times G_n$ defined by

$$\varphi_{\pi^{-1}}(g_1, g_2, \dots, g_n) = (g_{\pi(1)}, g_{\pi(2)}, \dots, g_{\pi(n)})$$

Then $\varphi_{\pi^{-1}}$ is a homomorphism by the same argument as above. Moreover, $\varphi_{\pi^{-1}}$ is the inverse of φ_π , since

$$\varphi_\pi(\varphi_{\pi^{-1}}(g_1, g_2, \dots, g_n)) = \varphi_\pi(g_{\pi(1)}, g_{\pi(2)}, \dots, g_{\pi(n)}) = (g_1, g_2, \dots, g_n)$$

and

$$\varphi_{\pi^{-1}}(\varphi_\pi(g_1, g_2, \dots, g_n)) = \varphi_{\pi^{-1}}(g_{\pi^{-1}(1)}, g_{\pi^{-1}(2)}, \dots, g_{\pi^{-1}(n)}) = (g_1, g_2, \dots, g_n).$$

Therefore, φ_π is an isomorphism. ■

Exercise 5.1.8

Let $G_1 = G_2 = \cdots = G_n$ and let $G = G_1 \times \cdots \times G_n$. Under the notation of the preceding exercise show that $\varphi_\pi \in \text{Aut}(G)$. Show also that the map $\pi \mapsto \varphi_\pi$ is an injective homomorphism of S_n into $\text{Aut}(G)$. (In particular, $\varphi_{\pi_1} \circ \varphi_{\pi_2} = \varphi_{\pi_1\pi_2}$. It is at this point that the π^{-1} 's in the definition of φ_π are needed. The underlying reason for this is because if e_i is the n -tuple with 1 in position i and zeros elsewhere, $1 \leq i \leq n$, then S_n acts on $\{e_1, \dots, e_n\}$ by $\pi \cdot e_i = e_{\pi(i)}$; this is a left group action. If the n -tuple $(g_1, \dots, g_n)$ is represented by $g_1e_1 + \cdots + g_ne_n$, then this left group action on $\{e_1, \dots, e_n\}$ extends to a left group action on sums by

$$\pi \cdot (g_1e_1 + g_2e_2 + \cdots + g_ne_n) = g_1e_{\pi(1)} + g_2e_{\pi(2)} + \cdots + g_ne_{\pi(n)}.$$

The coefficient of $e_{\pi(i)}$ on the right hand side is g_i , so the coefficient of e_i is $g_{\pi^{-1}(i)}$. Thus the right hand side may be rewritten as $g_{\pi^{-1}(1)}e_1 + g_{\pi^{-1}(2)}e_2 + \cdots + g_{\pi^{-1}(n)}e_n$, which is precisely the sum attached to the n -tuple $(g_{\pi^{-1}(1)}, g_{\pi^{-1}(2)}, \dots, g_{\pi^{-1}(n)})$. In other words, any permutation of the “position vectors” $e_1, \dots, e_n$ (which fixes their coefficients) is the same as the inverse permutation on the coefficients (fixing the e_i 's). If one uses π 's in place of π^{-1} 's in the definition of φ_π then the map $\pi \mapsto \varphi_\pi$ is not necessarily a homomorphism — it corresponds to a *right* group action.)

Solution. Note that $\varphi_\pi \in \text{Aut}(G)$ since φ_π is an isomorphism from G to itself. Let $\pi_1, \pi_2 \in S_n$. Then

$$\begin{aligned} \varphi_{\pi_1}(\varphi_{\pi_2}(g_1, g_2, \dots, g_n)) &= \varphi_{\pi_1}(g_{\pi_2^{-1}(1)}, g_{\pi_2^{-1}(2)}, \dots, g_{\pi_2^{-1}(n)}) \\ &= (g_{\pi_2^{-1}(\pi_1^{-1}(1))}, g_{\pi_2^{-1}(\pi_1^{-1}(2))}, \dots, g_{\pi_2^{-1}(\pi_1^{-1}(n))}) \\ &= (g_{(\pi_1\pi_2)^{-1}(1)}, g_{(\pi_1\pi_2)^{-1}(2)}, \dots, g_{(\pi_1\pi_2)^{-1}(n)}) \\ &= \varphi_{\pi_1\pi_2}(g_1, g_2, \dots, g_n) \end{aligned}$$

Therefore, the map $\pi \mapsto \varphi_\pi$ is a homomorphism. To see that it is injective, note that if φ_π is the identity automorphism, then for all i , $g_{\pi^{-1}(i)} = g_i$ for all $g_i \in G_i$, which implies π is the identity permutation. Hence, the map is injective. ■

Exercise 5.1.9

Let G_i be a field F for all i and use the preceding exercise to show that the set of $n \times n$ matrices with one 1 in each row and each column is a subgroup of $\text{GL}_n(F)$ isomorphic to S_n (these matrices are the *permutation matrices* since they simply permute the standard basis $e_1, \dots, e_n$ (as above) of the n -dimensional vector space $F \times F \times \dots \times F$).

Solution. Let $G = F \times F \times \dots \times F$ (n times) and let $\pi \in S_n$. Then φ_π is an automorphism of G that permutes the standard basis vectors $e_1, e_2, \dots, e_n$. In particular, $\varphi_\pi(e_i) = e_{\pi(i)}$ for all i . Thus, the matrix representation of φ_π with respect to the standard basis is a permutation matrix. Since the map $\pi \mapsto \varphi_\pi$ is an injective homomorphism from S_n to $\text{Aut}(G)$, the set of permutation matrices forms a subgroup of $\text{GL}_n(F)$ that is isomorphic to S_n . ■

Exercise 5.1.10

Let p be a prime. Let A and B be two cyclic groups of order p with generators x and y respectively. Set $E = A \times B$ so that E is the elementary abelian group of order p^2 : E_{p^2} . Prove that the distinct subgroups of E of order p are

$$\langle x \rangle, \quad \langle xy \rangle, \quad \langle xy^2 \rangle, \quad \dots, \quad \langle xy^{p-1} \rangle, \quad \langle y \rangle$$

(note that there are $p + 1$ of them).

Solution. Note that any proper, nontrivial subgroup of E must have order p , and every element of E has order p . The text shows that there are $p + 1$ subgroups of E of order p , so it suffices to show that the subgroups listed in the problem statement are distinct. Note that $\langle x \rangle$ and $\langle y \rangle$ are distinct since they have different generators. Moreover, $\langle xy^i \rangle$ is distinct from $\langle x \rangle$ and $\langle y \rangle$ for all i , since xy^i is not a power of x or y (as can be seen because $\langle x \rangle \cap \langle y \rangle$ is trivial). Finally, $\langle xy^i \rangle$ is distinct from $\langle xy^j \rangle$ for $i \neq j$, since if $\langle xy^i \rangle = \langle xy^j \rangle$, then xy^i is a power of xy^j , which implies y^{i-j} is a power of x , which is impossible since $\langle x \rangle \cap \langle y \rangle$ is trivial. Thus, all the subgroups listed in the problem statement are distinct. ■

Exercise 5.1.11

Let p be a prime, and let $n \in \mathbb{Z}^+$. Find a formula for the number of subgroups of order p in the elementary abelian group E_{p^n} .

Solution. Every element of E_{p^n} has order p , so every subgroup of order p is generated by a single non-identity element. Since there are $p^n - 1$ non-identity elements in E_{p^n} , and each subgroup of order p has $p - 1$ generators (the non-identity elements of the subgroup), the number of subgroups of order p in E_{p^n} is

$$\frac{p^n - 1}{p - 1} = 1 + p + p^2 + \dots + p^{n-1}$$

as desired. ■

(*) **Exercise 5.1.12**

Let A and B be groups. Assume $Z(A)$ contains a subgroup Z_1 and $Z(B)$ contains a subgroup Z_2 with $Z_1 \cong Z_2$. Let this isomorphism be given by the map $x_i \mapsto y_i$ for all $x_i \in Z_1$. A **central product** of A and B is a quotient

$$(A \times B)/Z \quad \text{where} \quad Z = \{(x_i, y_i^{-1}) \mid x_i \in Z_1\}$$

and is denoted by $A * B$ — it is not unique since it depends on Z_1, Z_2 and the isomorphism between them. (Think of $A * B$ as the direct product of A and B “collapsed” by identifying each element $x_i \in Z_1$ with its corresponding element $y_i \in Z_2$.)

- (a) Prove that the images of A and B in the quotient group $A * B$ are isomorphic to A and B , respectively, and that these images intersect in a central subgroup isomorphic to Z_1 . Find $|A * B|$.
- (b) Let $Z_4 = \langle x \rangle$. Let $D_8 = \langle r, s \rangle$ and $Q_8 = \langle i, j \rangle$ be given by their usual generators and relations. Let $Z_4 * D_8$ be the central product of Z_4 and D_8 which identifies x^2 and r^2 (i.e., $Z_1 = \langle x^2 \rangle$, $Z_2 = \langle r^2 \rangle$ and the isomorphism is $x^2 \mapsto r^2$) and let $Z_4 * Q_8$ be the central product of Z_4 and Q_8 which identifies x^2 and -1 . Prove that $Z_4 * D_8 \cong Z_4 * Q_8$.

Solution.

- (a) Consider the maps $\pi_A : A \rightarrow A * B$ and $\pi_B : B \rightarrow A * B$ defined by $\pi_A(a) = (a, 1)Z$ and $\pi_B(b) = (1, b)Z$. It is clear that π_A and π_B are homomorphisms. Moreover, $\ker \pi_A$ consists of the elements $a \in A$ such that $(a, 1) \in Z$. Then $y_i = 1$, and since $x_i \mapsto y_i$ is an isomorphism, we have $x_i = 1$, so $a = 1$. Thus, $\ker \pi_A$ is trivial, and π_A is injective. Similarly, $\ker \pi_B$ is trivial, and π_B is injective. Therefore, the images of A and B in $A * B$ are isomorphic to A and B , respectively. Moreover, the images of A and B in $A * B$ intersect in the subgroup $\{(x_i, 1)Z \mid x_i \in Z_1\}$, which is isomorphic to Z_1 . Finally, since Z has $|Z_1|$ elements, we have

$$|A * B| = \frac{|A||B|}{|Z_1|}.$$

- (b) Let Z_D denote the Z in the quotient for $Z_4 * D_8$ and let Z_Q denote the Z in the quotient for $Z_4 * Q_8$. Let x generate Z_4 in $Z_4 * D_8$ and y generate Z_4 in $Z_4 * Q_8$. Let r and s be the generators of D_8 in $Z_4 * D_8$ and let i and j be the generators of Q_8 in $Z_4 * Q_8$. Let $\mathbf{x}$ denote the coset $(x, 1)Z_D$ in $Z_4 * D_8$, and use similar notation for the elements r, s, y, i, j . Let $\mathbf{z} = \mathbf{j}\mathbf{y}^{-1}$. Consider the map $\varphi : Z_4 * D_8 \rightarrow Z_4 * Q_8$ defined by

$$\varphi(\mathbf{x}^\alpha \mathbf{r}^\beta \mathbf{s}^\gamma) = \mathbf{y}^\alpha \mathbf{i}^\beta \mathbf{z}^\gamma.$$

To see that this is well-defined, suppose $\mathbf{x}^\alpha \mathbf{r}^\beta \mathbf{s}^\gamma = \mathbf{1}$ in $Z_4 * D_8$. Then $\mathbf{r}^\beta \mathbf{s}^\gamma$ is in the image of Z_4 in $Z_4 * D_8$, so $\mathbf{r}^\beta \mathbf{s}^\gamma = \mathbf{x}^{2\delta}$ for some δ . Since r^2 is central in D_8 , we have $\mathbf{r}^\beta \mathbf{s}^\gamma = \mathbf{r}^{\beta+2\delta} \mathbf{s}^\gamma$. If $\gamma = 0$, then $\mathbf{r}^{\beta+2\delta} = \mathbf{1}$, so $4 \mid \beta + 2\delta$. If $\gamma = 1$, then $\mathbf{r}^{\beta+2\delta} \mathbf{s} = \mathbf{1}$, so $4 \mid \beta + 2\delta - 2$. In either case, we have $4 \mid \beta + 2\delta$, so $\mathbf{r}^\beta \mathbf{s}^\gamma = \mathbf{x}^{2\delta} = \mathbf{1}$. Thus, α is even, and we have $\varphi(\mathbf{x}^\alpha \mathbf{r}^\beta \mathbf{s}^\gamma) = \mathbf{y}^\alpha \mathbf{i}^\beta \mathbf{z}^\gamma = \mathbf{1}$. Therefore, φ is well-defined. It is clear that φ is a homomorphism. Moreover, if $\varphi(\mathbf{x}^\alpha \mathbf{r}^\beta \mathbf{s}^\gamma) = \mathbf{1}$, then α is even and β and γ are zero, so φ is injective. Finally, since $|Z_4 * D_8| = |Z_4 * Q_8|$, we conclude that φ is an isomorphism. ■

Exercise 5.1.13

Give presentations for the groups $Z_4 * D_8$ and $Z_4 * Q_8$ constructed in the preceding exercise.

Solution. To construct the presentations, we note that $\mathbf{x}$ should satisfy the relations of Z_4 , $\mathbf{r}$ and $\mathbf{s}$ should satisfy the relations of D_8 , and $\mathbf{i}$ and $\mathbf{z}$ should satisfy the relations of Q_8 . Moreover, we have the relations $\mathbf{x}^2 = \mathbf{r}^2$ in $Z_4 * D_8$, since $\mathbf{x}^2 \mathbf{r}^{-2} = \mathbf{1}$. Lastly, $\mathbf{x}$ is central in that $\mathbf{x}\mathbf{r} = \mathbf{r}\mathbf{x}$ and $\mathbf{x}\mathbf{s} = \mathbf{s}\mathbf{x}$ in $Z_4 * D_8$. Then a presentation for $Z_4 * D_8$ is

$$\langle \mathbf{x}, \mathbf{r}, \mathbf{s} \mid \mathbf{r}^4 = \mathbf{s}^2 = \mathbf{1}, \mathbf{x}^2 = \mathbf{r}^2, \mathbf{s}\mathbf{r}\mathbf{s} = \mathbf{r}^{-1}, \mathbf{x}\mathbf{r} = \mathbf{r}\mathbf{x}, \mathbf{x}\mathbf{s} = \mathbf{s}\mathbf{x} \rangle.$$

To find a presentation for $Z_4 * Q_8$, we may use the isomorphism φ from the preceding exercise to rewrite the presentation for $Z_4 * D_8$ in terms of $\mathbf{y}, \mathbf{i}, \mathbf{z}$. In particular, we have $\mathbf{y}^2 = \mathbf{i}^2 \mathbf{z}^2$, $\mathbf{z}^2 = \mathbf{1}$, and $\mathbf{y}\mathbf{i} = \mathbf{i}\mathbf{y}\mathbf{z}$. Thus, a presentation for $Z_4 * Q_8$ is

$$\langle \mathbf{y}, \mathbf{i}, \mathbf{z} \mid \mathbf{i}^4 = \mathbf{z}^2 = \mathbf{1}, \mathbf{y}^2 = \mathbf{i}^2, \mathbf{z}\mathbf{i}\mathbf{z} = \mathbf{i}^{-1}, \mathbf{y}\mathbf{i} = \mathbf{i}\mathbf{y}, \mathbf{y}\mathbf{z} = \mathbf{z}\mathbf{y} \rangle. \quad \blacksquare$$

Exercise 5.1.14

Let $G = A_1 \times A_2 \times \cdots \times A_n$, and for each i , let B_i be a normal subgroup of A_i . Prove that $B_1 \times B_2 \times \cdots \times B_n \trianglelefteq G$, and that

$$(A_1 \times A_2 \times \cdots \times A_n) / (B_1 \times B_2 \times \cdots \times B_n) \cong (A_1/B_1) \times (A_2/B_2) \times \cdots \times (A_n/B_n).$$

Solution. Let $G = A_1 \times A_2 \times \cdots \times A_n$ and let $B_i \trianglelefteq A_i$ for all i . Let $H = B_1 \times B_2 \times \cdots \times B_n$. It is clear that H is a subgroup of G . Moreover, since each B_i is normal in A_i , we have that for any $g = (a_1, a_2, \dots, a_n) \in G$ and any $h = (b_1, b_2, \dots, b_n) \in H$, we have

$$ghg^{-1} = (a_1 b_1 a_1^{-1}, a_2 b_2 a_2^{-1}, \dots, a_n b_n a_n^{-1}) \in H$$

since $a_i b_i a_i^{-1} \in B_i$ for all i . Thus, H is normal in G .

Consider the map $\varphi : G \rightarrow (A_1/B_1) \times (A_2/B_2) \times \cdots \times (A_n/B_n)$ defined by

$$\varphi((a_1, a_2, \dots, a_n)) = (a_1 B_1, a_2 B_2, \dots, a_n B_n).$$

It is clear that φ is a homomorphism. Moreover, φ is surjective since for any $(a_1 B_1, a_2 B_2, \dots, a_n B_n)$, we have $\varphi((a_1, a_2, \dots, a_n)) = (a_1 B_1, a_2 B_2, \dots, a_n B_n)$. Finally, the kernel of φ is the set of all $(a_1, a_2, \dots, a_n)$ such that $a_i \in B_i$ for all i , which is exactly H . Thus, by the First Isomorphism Theorem, we have

$$G/H \cong (A_1/B_1) \times (A_2/B_2) \times \cdots \times (A_n/B_n)$$

as desired. ■

Exercise 5.1.15

Let I be any nonempty index set and let $(G_i, \star_i)$ be a group for each $i \in I$. The *direct product* of the groups $G_i, i \in I$ is the set $G = \prod_{i \in I} G_i$ (the Cartesian product of the G_i 's) with a binary operation defined as follows: if $\prod a_i$ and $\prod b_i$ are elements of G , then their product in G is given by

$$\left(\prod_{i \in I} a_i \right) \left(\prod_{i \in I} b_i \right) = \prod_{i \in I} (a_i \star_i b_i)$$

(i.e., the group operation in the direct product is defined componentwise).

- (a) Show that this binary operation is well defined and associative.
- (b) Show that the element $\prod 1_i$ satisfies the axiom for the identity of G , where 1_i is the identity of G_i for all i .
- (c) Show that the element $\prod a_i^{-1}$ is the inverse of $\prod a_i$, where the inverse of each component element a_i is taken in the group G_i .

Conclude that the direct product is a group. (Note that if $I = \{1, 2, \dots, n\}$, this definition of the direct product is the same as the n -tuple definition in the text.)

Solution.

- (a) To show well-definedness, let $\prod \alpha_i = \prod a_i$ and $\prod \beta_i = \prod b_i$. Then $\alpha_i = a_i$ and $\beta_i = b_i$ for all i , so

$$\left(\prod_{i \in I} \alpha_i \right) \left(\prod_{i \in I} \beta_i \right) = \prod_{i \in I} (\alpha_i \star_i \beta_i) = \prod_{i \in I} (a_i \star_i b_i)$$

as desired. To show associativity, let $\prod a_i$, $\prod b_i$, and $\prod c_i$ be elements of G . Then

$$\begin{aligned} \left(\prod_{i \in I} a_i \right) \left(\left(\prod_{i \in I} b_i \right) \left(\prod_{i \in I} c_i \right) \right) &= \left(\prod_{i \in I} a_i \right) \left(\prod_{i \in I} (b_i \star_i c_i) \right) \\ &= \prod_{i \in I} (a_i \star_i (b_i \star_i c_i)) \\ &= \prod_{i \in I} ((a_i \star_i b_i) \star_i c_i) \\ &= \left(\prod_{i \in I} (a_i \star_i b_i) \right) \left(\prod_{i \in I} c_i \right) \\ &= \left(\left(\prod_{i \in I} a_i \right) \left(\prod_{i \in I} b_i \right) \right) \left(\prod_{i \in I} c_i \right) \end{aligned}$$

as desired.

(b) Let $\prod a_i$ be an element of G . Then

$$\left(\prod_{i \in I} 1_i \right) \left(\prod_{i \in I} a_i \right) = \prod_{i \in I} (1_i \star_i a_i) = \prod_{i \in I} a_i$$

and

$$\left(\prod_{i \in I} a_i \right) \left(\prod_{i \in I} 1_i \right) = \prod_{i \in I} (a_i \star_i 1_i) = \prod_{i \in I} a_i.$$

Thus, $\prod 1_i$ is the identity element of G .

(c) Let $\prod a_i$ be an element of G . Then

$$\left(\prod_{i \in I} a_i \right) \left(\prod_{i \in I} a_i^{-1} \right) = \prod_{i \in I} (a_i \star_i a_i^{-1}) = \prod_{i \in I} 1_i$$

and

$$\left(\prod_{i \in I} a_i^{-1} \right) \left(\prod_{i \in I} a_i \right) = \prod_{i \in I} (a_i^{-1} \star_i a_i) = \prod_{i \in I} 1_i.$$

Thus, $\prod a_i^{-1}$ is the inverse of $\prod a_i$. We conclude that G is a group with inverse and identity as described above. ■

Exercise 5.1.16

State and prove the generalization of Proposition 2 to arbitrary direct products.

Solution. Proposition 2 stated for direct arbitrary groups is as follows: if $G = \prod_{i \in I} G_i$ is a direct product of groups, then

1. Fix each i with j -th coordinate where $j \neq i$ is the identity element of G_j for $1 \leq j \leq n$. Then

$$\widehat{G}_i = \{(1, 1, \dots, 1, g_i, 1, \dots) \mid g_i \in G_i\}$$

is a normal subgroup of G that is isomorphic to G_i , or that

$$G/\widehat{G}_i \cong \prod_{j \neq i} G_j.$$

2. For fixed i , define $\pi_i : G \rightarrow G_i$ by $\pi_i((g_j)_{j \in I}) = g_i$. Then π_i is a surjective homomorphism with kernel $\widehat{G}_i$, so $G/\widehat{G}_i \cong G_i$.

3. If $x \in G_i$ and $y \in G_j$ for $i \neq j$, then x and y commute in G .

The proof is similar to the proof of Proposition 2 in the text.

1. It is clear that $\widehat{G}_i$ is a subgroup of G . Moreover, for any $\prod_{j \in I} g_j \in G$ and any $h = \prod_{j \neq i} 1_j h_i \in \widehat{G}_i$, we have

$$\left(\prod_{j \in I} g_j \right) h \left(\prod_{j \in I} g_j \right)^{-1} = \prod_{j \neq i} 1_j (g_i h_i g_i^{-1}) \in \widehat{G}_i$$

since $g_i h_i g_i^{-1} \in G_i$. Thus, $\widehat{G}_i$ is normal in G . Consider the map $\varphi : G_i \rightarrow \widehat{G}_i$ defined by $\varphi(g_i) = \prod_{j \neq i} 1_j g_i$. It is clear that φ is a homomorphism. Moreover, if $\varphi(g_i) = \prod_{j \neq i} 1_j g_i = \prod_{j \neq i} 1_j$, then g_i is the identity element of G_i , so φ is injective. Finally, for any $\prod_{j \neq i} 1_j h_i \in \widehat{G}_i$, we have $\varphi(h_i) = \prod_{j \neq i} 1_j h_i$, so φ is surjective. Thus, $\widehat{G}_i$ is isomorphic to G_i .

Consider the map

$$\psi : G \rightarrow \prod_{j \neq i} G_j \text{ defined by } \psi \left(\prod_{j \in I} g_j \right) = \prod_{j \neq i} g_j.$$

It is clear that ψ is a homomorphism. Moreover, the kernel of ψ is $\widehat{G}_i$, so by the First Isomorphism Theorem, we have

$$G/\widehat{G}_i \cong \prod_{j \neq i} G_j.$$

2. For fixed i , it is clear that π_i is a homomorphism. Moreover, π_i is surjective since for any $g_i \in G_i$, we have $\pi_i(\prod_{j \in I} 1_j g_i) = g_i$. Finally, the kernel of π_i is $\widehat{G}_i$, so by the First Isomorphism Theorem, we have

$$G/\widehat{G}_i \cong G_i.$$

3. Let $x = \prod_{j \neq i} 1_j x_i$ and $y = \prod_{j \neq k} 1_j y_k$ for some $i \neq k$. Then

$$xy = \prod_{j \neq i, k} 1_j x_i y_k = \prod_{j \neq i, k} 1_j y_k x_i = yx$$

as desired. ■

Exercise 5.1.17

Let I be any nonempty index set and let G_i be a group for each $i \in I$. The *restricted direct product* or *direct sum* of the groups G_i is the set of elements of the direct product which are the identity in all but finitely many components, that is, the set of all elements $\prod a_i \in \prod_{i \in I} G_i$ such that $a_i = 1_i$ for all but a finite number of $i \in I$.

- Prove that the restricted direct product is a subgroup of the direct product.
- Prove that the restricted direct product is normal in the direct product.
- Let $I = \mathbb{Z}^+$ and let p_i be the i th integer prime. Show that if $G_i = \mathbb{Z}/p_i\mathbb{Z}$ for all $i \in \mathbb{Z}^+$, then every element of the restricted direct product of the G_i 's has finite order but $\prod_{i \in \mathbb{Z}^+} G_i$ has elements of infinite order. Show that in this example the restricted direct product is the torsion subgroup of the direct product (cf. Exercise 2.1.16).

Solution.

- Let $G = \prod_{i \in I} G_i$ and let H be the restricted direct product of the G_i 's. It is clear that H is a subset of G . Moreover, if $\prod a_i$ and $\prod b_i$ are elements of H , then there are only finitely many i such that $a_i \neq 1_i$ and only finitely many i such that $b_i \neq 1_i$. Thus, there are only finitely many i such that $a_i b_i \neq 1_i$, so $\prod a_i \prod b_i = \prod (a_i b_i)$ is an element of H . Finally, if $\prod a_i$ is an element of H , then there are only finitely many i such that $a_i \neq 1_i$, so there are only finitely many i such that $a_i^{-1} \neq 1_i$, so $\prod a_i^{-1}$ is an element of H . Thus, $H \leq G$.

- (b) Let $G = \prod_{i \in I} G_i$ and let H be the restricted direct product of the G_i 's. Let $\prod a_i$ be an element of G and let $\prod b_i$ be an element of H . Then there are only finitely many i such that $b_i \neq 1_i$, so there are only finitely many i such that $a_i b_i a_i^{-1} \neq 1_i$, so $\prod a_i \prod b_i \prod a_i^{-1}$ is an element of H . Thus, $H \trianglelefteq G$.
- (c) Let $G = \prod_{i \in \mathbb{Z}^+} G_i$ where $G_i = \mathbb{Z}/p_i \mathbb{Z}$ for all i . Let H be the restricted direct product of the G_i 's. If $\prod a_i$ is an element of H , then there are only finitely many i such that $a_i \neq 1_i$, so there are only finitely many i such that a_i has order greater than 1, so $\prod a_i$ has finite order. On the other hand, if $\prod a_i$ is an element of G such that a_i is a generator of G_i for all i , then $\prod a_i$ has infinite order since for any positive integer n , we have $(\prod a_i)^n = \prod a_i^n$, and there are infinitely many i such that $a_i^n \neq 1_i$.

To show $H = \text{Tor}(G)$, we note that $H \subseteq \text{Tor}(G)$ since every element of H has finite order. Conversely, if $\prod a_i$ is an element of $\text{Tor}(G)$, then there is some positive integer n such that $(\prod a_i)^n = \prod a_i^n = \prod 1_i$, so $a_i^n = 1_i$ for all i . Since G_i is cyclic of prime order, this implies that a_i is either the identity or a generator of G_i . If there are infinitely many i such that a_i is a generator of G_i , then $\prod a_i$ has infinite order, which is a contradiction. Thus, there are only finitely many i such that a_i is a generator of G_i , so there are only finitely many i such that $a_i \neq 1_i$, so $\prod a_i$ is an element of H . Therefore, $\text{Tor}(G) \subseteq H$, and we conclude that $\text{Tor}(G) = H$. ■

Exercise 5.1.18

In each of (a) to (e) give an example of a group with the specified properties:

- (a) an infinite group in which every element has order 1 or 2
- (b) an infinite group in which every element has finite order but for each positive integer n there is an element of order n
- (c) a group with an element of infinite order and an element of order 2
- (d) a group G such that every finite group is isomorphic to some subgroup of G
- (e) a nontrivial group G such that $G \cong G \times G$.

Solution.

- (a) Consider the infinite direct product $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/2\mathbb{Z} \times \cdots$. Every element of this group is a tuple of 0's and 1's, and the order of each element is either 1 (if the tuple is the identity) or 2 (if the tuple has at least one 1).
- (b) Let $G = \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z} \times \mathbb{Z}/4\mathbb{Z} \times \cdots$. Every element of G has finite order since it is a tuple of elements from finite cyclic groups. Moreover, for each positive integer n , the element that is the identity in all components except the n -th component, which is a generator of $\mathbb{Z}/n\mathbb{Z}$, has order n .
- (c) Consider the group $\mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}$. The element $(1, 0)$ has order 2, and the element $(0, 1)$ has infinite order.
- (d) By Cayley's Theorem, every finite group is isomorphic to a subgroup of some symmetric group S_n . We may then consider the group $G = S_1 \times S_2 \times S_3 \times \cdots$. For any finite group H , there is some n such that H is isomorphic to a subgroup of S_n , so H is isomorphic to a subgroup of G .
- (e) Let $G = \mathbb{Z} \times \mathbb{Z} \times \cdots$. Consider the map $\varphi : G \rightarrow G \times G$ defined by

$$\varphi((a_1, a_2, a_3, \dots)) = ((a_1, a_3, a_5, \dots), (a_2, a_4, a_6, \dots)).$$

It is clear that φ is a homomorphism. Moreover, if $\varphi((a_1, a_2, a_3, \dots)) = ((0, 0, 0, \dots), (0, 0, 0, \dots))$, then $a_i = 0$ for all i , so φ is injective. Finally, let $((b_1, b_2, b_3, \dots), (c_1, c_2, c_3, \dots))$ be an element of $G \times G$. Then $\varphi((b_1, c_1, b_2, c_2, b_3, c_3, \dots)) = ((b_1, b_2, b_3, \dots), (c_1, c_2, c_3, \dots))$, so φ is surjective. Thus, φ is an isomorphism and $G \cong G \times G$. ■